SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

Technical Presentation – Assessment Tool 3

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CSA5103 – CRYPTOGRAPHY AND NETWORK SECURITY

CO1: Classical Encryption Techniques (BL3, BL4)

CO1
AT – 3

TOPIC 1

From Caesar to Vigenère:
Evolution of Substitution Ciphers

TOPIC 2

Role of Number Theory in Cryptography: From Euclid to Euler

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SSE, SIMATS

Let's decode!



SECURE
★ CSA51 ★
ENCRYPTED

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AGENDA & OBJECTIVES

1

TOPIC 1 · Substitution Ciphers

- Caesar cipher and its weaknesses
- Evolution: monoalphabetic → polyalphabetic
- Vigenère cipher: working with an example
- Comparison and cryptanalysis

2

TOPIC 2 · Number Theory

- Euclid: gcd and modular inverse
- Fermat and Euler: φ(n) and the theorems
- RSA worked example with Python check
- Modern applications and security basis

Learning objectives (CO1)

- Apply classical ciphers to encrypt and decrypt messages.
- Analyse why classical ciphers fail and how mathematics fixes it.
- Verify every result by hand and by a short Python program.



CAESAR & THE BIRTH OF SUBSTITUTION CIPHERS

- **Substitution cipher:** each plaintext letter is replaced by another symbol; the order of letters does not change.
- **Caesar cipher:** every letter is shifted by a fixed key k (Julius Caesar used k = 3).
- **Encrypt:** `C = (P + k) mod 26`
- **Decrypt:** `P = (C − k) mod 26`
- **Weakness 1:** only 25 useful keys, so brute force takes seconds.
- **Weakness 2:** letter frequencies are preserved (E stays the commonest letter), so frequency analysis works.

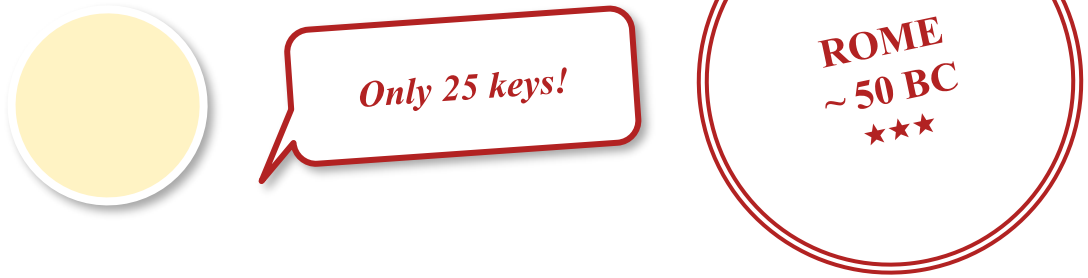
Plain	A	B	C	D	E	F	G	H	I	J	K	L	M
Cipher	**D**	**E**	**F**	**G**	**H**	**I**	**J**	**K**	**L**	**M**	**N**	**O**	**P**

Plain	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher	**Q**	**R**	**S**	**T**	**U**	**V**	**W**	**X**	**Y**	**Z**	**A**	**B**	**C**

Cipher alphabet for shift k = 3

HELLO → KHOOR

plaintext → ciphertext (k = 3)



EVOLUTION OF SUBSTITUTION CIPHERS

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MONO → POLY

~50 BC	9th century	1553	1854	1863	
1	2	3	4	5	6
Caesar cipher	**Al-Kindi**	**Bellaso / Vigenère**	**Playfair**	**Kasiski test**	**Vernam, rotor machines**
One fixed shift; only 25 keys	First frequency analysis breaks monoalphabetic ciphers	Keyword gives many shifts: polyalphabetic	Encrypts pairs of letters (digraphs)	Finds the Vigenère key length	Random key (one-time pad); Enigma

- Monoalphabetic
- Polyalphabetic / stronger
- Cryptanalysis (attack)

Every attack forced a stronger cipher

VIGENÈRE CIPHER IN ACTION

Step	1	2	3	4	5
Plaintext	**H**	**E**	**L**	**L**	**O**
P value (A=0)	7	4	11	11	14
Repeating key (KEY)	**K**	**E**	**Y**	**K**	**E**
K value	10	4	24	10	4
P + K	17	8	35	21	18
(P + K) mod 26	17	8	9	21	18
Ciphertext	**R**	**I**	**J**	**V**	**S**

Encrypt: $C = (P + K) \bmod 26$

Decrypt: $P = (C - K) \bmod 26$

K repeats: K, E, Y, K, E ...

Why it is stronger

The two L's become J and V: the same letter gets different cipher letters.

Python check ✓

```
def vig(p, k):
    return "".join(chr((ord(c) + ord(k[i % len(k)]) - 130) % 26 + 65)
                   for i, c in enumerate(p))
print(vig("HELLO", "KEY"))   # RIJVS
```

HELLO → RIJVS

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CAESAR vs VIGENÈRE: SECURITY ANALYSIS

Feature	Caesar	Vigenère
Key	One shift k (0–25)	Keyword of length m
Key space	25 keys	26^m (m = 5 → 11,881,376)
Letter frequencies	Preserved: peaks stay visible	Flattened across m alphabets
Typical attack	Brute force, frequency analysis	Kasiski / index of coincidence, then column-wise frequency analysis
Security today	**None**	**Broken (teaching use only)**

Lesson: a short repeating key always leaks its length.

Breaking Vigenère

1. Find key length m (Kasiski or index of coincidence)
2. Split the text into m columns
3. Run frequency analysis on each column

BROKEN
KASISKI
1863

EUCLID: GCD & MODULAR INVERSE

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- **Euclid's algorithm:** gcd(a, b) = gcd(b, a mod b); repeat until the remainder is 0.
- **Extended Euclid:** writes gcd = a·x + b·y, giving the inverse a^{-1} mod n when gcd = 1.
- **Affine cipher:** C = (aP + b) mod 26 needs gcd(a, 26) = 1, which leaves 12 values of a and 12 × 26 = 312 keys.
- **RSA:** the private key d is found with extended Euclid.

c. 300 BC
Euclid's
Elements

gcd = 1
✓



Example: find 7^{-1} mod 40

Forward (divide)	Backward (isolate remainder)
40 = 5 × 7 + 5	5 = 40 − 5×7
7 = 1 × 5 + 2	2 = 7 − 1×5
5 = 2 × 2 + 1	1 = 5 − 2×2

$1 = 3\times40 - 17\times7 \Rightarrow 7^{-1} \equiv -17 \equiv 23 \pmod{40}$

Check: 7 × 23 = 161 = 4×40 + 1 ✓

No factoring needed to find gcd!

FERMAT & EULER: THE KEY THEOREMS

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1640

FERMAT'S LITTLE THEOREM

If p is prime and p ∤ a:

a^(p−1) ≡ 1 (mod p)

Example: $3^6 = 729 = 104 \times 7 + 1$, so $3^6 \equiv 1 \pmod 7$

Use: fast primality tests and key generation

1763

EULER'S TOTIENT φ(n)

φ(n) = count of numbers 1..n coprime to n

$\varphi(p) = p - 1$

$\varphi(pq) = (p-1)(q-1)$

Example: $\varphi(15) = 2 \times 4 = 8$

1763

EULER'S THEOREM

If gcd(a, n) = 1:

a^φ(n) ≡ 1 (mod n)

Example: $2^8 = 256 = 17 \times 15 + 1$, so $2^8 \equiv 1 \pmod{15}$

Use: proves why RSA decryption works

Fermat's theorem is the special case of Euler's theorem with n = p, since φ(p) = p − 1.

$e \cdot d \equiv 1 \pmod{\varphi(n)}$

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RSA: NUMBER THEORY AT WORK



RSA 1977

Step	Computation
1	Pick primes p = 5, q = 11 ⇒ n = 55
2	φ(n) = (5−1)(11−1) = 40
3	Choose e = 7, since gcd(7, 40) = 1
4	$d = 7^{-1} \bmod 40 = 23$ (extended Euclid)
5	Public key (55, 7); private key d = 23
6	Encrypt m = 8: $8^7 \bmod 55 = 2$
7	**Decrypt: $2^{23} \bmod 55 = 8$ ✓**

Python verification

```
p, q = 5, 11
n, phi = p*q, (p-1)*(q-1)
e = 7
d = pow(e, -1, phi)    # 23
c = pow(8, e, n)       # 2
m = pow(c, d, n)       # 8
```

Security: an attacker sees n = 55 but must factor it to get φ(n). Real RSA uses n of 2048 bits.

ROUND TRIP
8 → 2 → 8



Euclid + Euler = RSA

NUMBER THEORY IN MODERN CRYPTOGRAPHY

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System	Number-theory tool	Used for	Security rests on
RSA	φ(n), modular inverse, modular exponentiation	Encryption, signatures	Factoring n = pq is hard
Diffie–Hellman	Primes, primitive roots, mod p powers	Key exchange (TLS)	Discrete logarithm problem
Elliptic curves (ECC)	Arithmetic on curves over mod p	Mobile, TLS, Bitcoin	Elliptic-curve discrete log
Miller–Rabin / Fermat test	Fermat's little theorem	Generating large primes	Fast probabilistic primality test
Caesar, Vigenère, affine	Only mod 26 arithmetic and gcd	Teaching, history	**Nothing hard: small key space**

From mod 26 to 2048-bit primes: same mathematics, much bigger numbers and genuinely hard problems.





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CONCLUSION & KEY TAKEAWAYS

Topic 1 · Substitution Ciphers

- Caesar uses one shift; Vigenère uses many, hiding letter frequencies.
- A short repeating key still leaks its length (Kasiski, index of coincidence).
- Lesson: security needs a huge key space and no patterns.

Topic 2 · Number Theory

KEY POINTS

- Euclid gives gcd and modular inverses.
- Fermat and Euler give φ(n) and the proof that RSA decryption works.
- Hard problems (factoring, discrete log) give modern security.

MISSION COMPLETE ★★★

References

- W. Stallings, Cryptography and Network Security, Pearson.
- D. Kahn, The Codebreakers, Scribner.
- A. Menezes, P. van Oorschot, S. Vanstone, Handbook of Applied Cryptography, CRC Press.

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THANK YOU



Questions & Discussion

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✓ DELIVERED
✓

WKDQN BRX = THANK YOU (Caesar, k = 3)

Any Questions ?